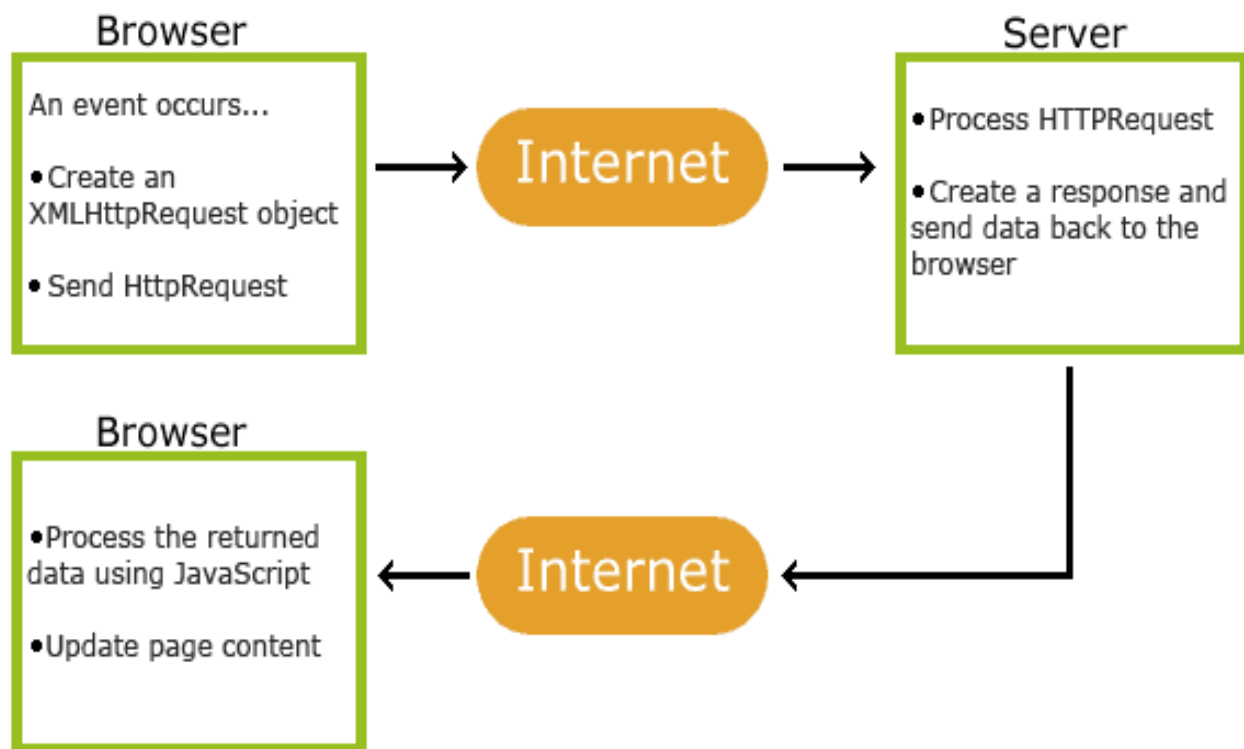


AJAX

Overview of AJAX

- AJAX is Asynchronous JavaScript and XML
- AJAX is user experience. With the help of AJAX we can communicate with server without page reload.
- AJAX allows web pages to be updated asynchronously by exchanging small amounts of data with web server behind the scene.
- This means that it is possible to update parts of a web page, without reloading the whole page.

How AJAX Works



Types of AJAX

- ❖ Asynchronous Ajax
 - ✎ Asynchronous Ajax call allow the next line of code to execute.
- ❖ Synchronous Ajax
 - ✎ Synchronous call stop JavaScript execution until the response from server.

AJAX Components

The key technical component of AJAX are:

- XHTML – a stricter, cleaner rendering of HTML into XML.

- CSS for marking up and adding styles.
- The JavaScript Document Object Model (DOM) which allows the content, structure and style of a document to be dynamically accessed and updated.
- The **XMLHttpRequest** object which exchanges data asynchronously with the Web server reducing the need to continually fetch resources from the server.

Asynchronous Data Transfer with XMLHttpRequest

- All modern browser support the XMLHttpRequest object.
- The XMLHttpRequest object is used to exchange data with a server behind the scenes.
- Req=new XMLHttpRequest();
- Sending request to the server

Req.open("GET","abc.php", true);

Req.send();

Req.open("GET","abc.php?x=25", true);

Req.send();

XMLHttpRequest Object Methods

Method	Description
new XMLHttpRequest()	Creates a new XMLHttpRequest object
abort()	Cancels the current request
getAllResponseHeaders()	Returns header information
getResponseHeader()	Returns specific header information
open(<i>method, url, async, user, psw</i>)	Specifies the request <i>method</i> : the request type GET or POST <i>url</i> : the file location <i>async</i> : true (asynchronous) or false (synchronous) <i>user</i> : optional user name <i>psw</i> : optional password
send()	Sends the request to the server Used for GET requests

send(<i>string</i>)	Sends the request to the server. Used for POST requests
-----------------------	--

XMLHttpRequest Object Properties

Property	Description
onreadystatechange	Defines a function to be called when the readyState property changes
readyState	Holds the status of the XMLHttpRequest. 0: request not initialized 1: server connection established 2: request received 3: processing request 4: request finished and response is ready
responseText	Returns the response data as a string
responseXML	Returns the response data as XML data
status	Returns the status-number of a request 200: "OK" 403: "Forbidden" 404: "Not Found" For a complete list go to the Http Messages Reference
statusText	Returns the status-text (e.g. "OK" or "Not Found")

Onreadystatechange

- When a request to a server is sent. We want to perform some actions based on the response.
- The onreadystatechange event is triggered every time the readyState changes.
- The onreadystatechange stores a function to be called automatically each time the readyState property change.

Code

```
JS{
```

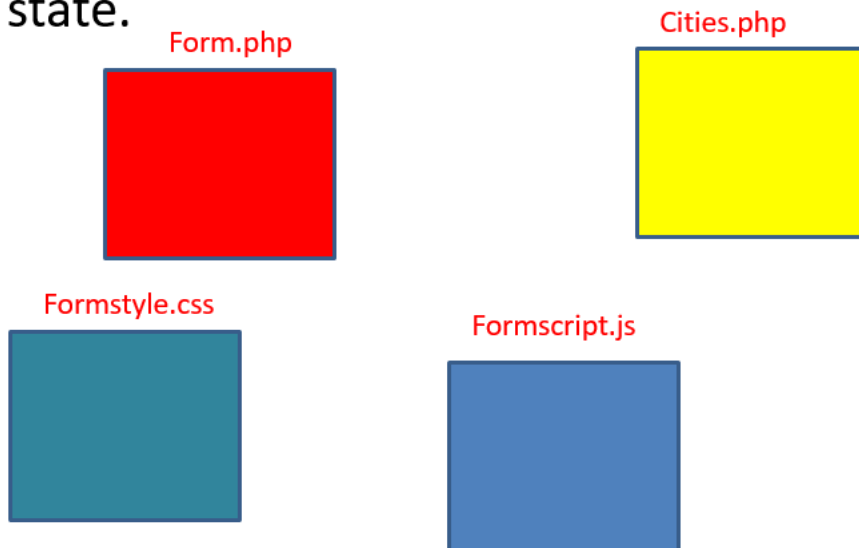
```

var req= new XMLHttpRequest();
    req.open("GET","abc.php", true);
    req.send();
    req.onreadystatechange=function()
{
    if(req.readyState==4 && req.status==200)
    {
        document.getElementById("myDiv").innerHTML;
        ML= req.responseText;
    }
}
}
}

```

Example

- Fetching city name from server on selection of state.



```

<!DOCTYPE html>
<html>
<head>
    <title>Ajax Example</title>
    <link rel="stylesheet" type="text/css" href="formstyle.css">
    <script type="text/javascript" src="formscript.js"></script>
</head>
<body>
    <div id="form-div">
        <h1>Application form</h1>
    
```

```

        <form action="#" method="post">
            <div>
                <label>Name</label>
                <input type="text" name="uname">
            </div>
            <div>
                <label>Select state</label>
                <select onchange="fetchcities(this.value)">
                    <option>Select state </option>
                    <option>Pradesh1</option>
                    <option>Pradesh3</option>
                    <option>Pradesh7</option>
                </select>
            </div>
            <div>
                <label>Select City</label>
                <select id="cities">
                    <option>First Select state </option>
                </select>
            </div>
            <div>
                <input type="submit" name="submit" value="submit">
            </div>
        </form>
    </div>
</body>
</html>

```

```

Formstyle.css
#form-div
{
    position: relative;
    left: 30%;
    width: 400px;
    background-color: grey;
    padding: 30px
}
#form-div div
{

```

```

        margin: 10px;
    }
    input,select{
        display: block;
        padding: 10px;
        width: 100%;
        border-radius: 4px;
    }

```

Formscript.js

```

function fetchcities(str)
{
    //alert(str);
    var req=new XMLHttpRequest();
    req.open("get","http://localhost:81/Ajax1/cities.php?state="+str,true);
    req.send();
    req.onreadystatechange=function(){
        if (req.readyState==4 && req.status==200) {

            document.getElementById("cities").innerHTML=req.responseText;
        }
    };
}

```

```

<?php
$state=$_GET['state'];
$c1=array('jhapa','dhankuta');
$c2=array('katmandu','lalitpur','Bhaktpur');
$c3=array('dhangadi','mahendranagar');
if ($state=='Pradesh1') {
    foreach ($c1 as $v) {
        echo "<option>$v</option>";
    }
}
else if ($state=='Pradesh3') {
    foreach ($c2 as $v) {
        echo "<option>$v</option>";
    }
}
else if($state=='Pradesh7')
{
    foreach ($c3 as $v) {

```

```
        echo "<option>$v</option>";
    }
} else
{
    echo "<option>select First State</option>";
}
?>
```